INCH-POUND
FF-W-92C
w/ AMENDMENT 2
2 May 2024
SUPERSEDING
FF-W-92C
w/ AMENDMENT 1
14 October 2022

FEDERAL SPECIFICATION
WASHER, FLAT (PLAIN)

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1 SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plain, flat washers.

1.2 Classification. Flat washers covered by this specification shall be of the following types, grades and classes, as specified (see 6.2).

Type:

A - Round washer
B - Square washer

Grade:

I - General assembly purpose washer
II - Aeronautical assembly purpose washer

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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Class:

A - Carbon steel washer
B - Corrosion-resisting steel washer
C - Aluminum washer
D - Aluminum alloy washer
E - Brass washer
F - Copper washer
G - Nickel-copper alloy washer
H - Plastic (nylon) washer
J - Nickel-copper aluminum alloy washer

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip Wire, Forgings and Structural and Special Shaped Sections
QQ-N-286	-	Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-13924	-	Coating, Oxide, Black, for Ferrous Metals
MIL-DTL-16232	-	Phosphate Coating, Heavy, Manganese or Zinc Base
MIL-DTL-495	-	Finish, Chemical, Black, for Copper Alloys
MIL-PRF-8625	-	Anodic Coatings for Aluminum and Aluminum Alloy
MIL-PRF-32647	-	Zinc-Nickel Electroplating for Fasteners

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available from https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.21.1	-	Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

(Copies of this document are available from www.asme.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes
ASQ H1607	-	Zero Acceptance Number Sampling Plans

(Copies of this document are available from www.asq.org.)

ASTM INTERNATIONAL

ASTM A153/A153M	-	Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
ASTM A342/A342M	-	Standard Test Methods for Permeability of Weakly Magnetic Materials
ASTM A967/A967M	-	Standard Specification for Chemical Passivation Treatments for Stainless Steel Parts
ASTM B36/B36M	-	Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B121/B121M	-	Standard Specification for Leaded Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B152/B152M	-	Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar
ASTM B209/B209M	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B339	-	Standard Specification for Pig Tin
ASTM B545	-	Standard Specification for Electrodeposited

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		Coatings of Tin
ASTM B633	-	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM D4066	-	Standard Classification System for Nylon Injection and Extrusion Materials (PA)

(Copies of these documents are available from www.astm.org.)

NATIONAL AEROSPACE STANDARD (NAS)

NAS1149	-	Washer, Flat

(Copies of this document are available from www.aia-aerospace.org.)

SAE INTERNATIONAL

SAE AMS2461	-	Plating, Zinc-Nickel Alloy (12 to 16% Ni)
SAE AMS2700	-	Passivation of Corrosion Resistant Steels
SAE AIR4127	-	Steel: Chemical Composition and Hardenability
SAE AMS-C-81562	-	Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited)
SAE AMS-QQ-A-250	-	Aluminum and Aluminum Alloy, Plate and Sheet General Specification for
SAE AMS-QQ-N-290	-	Nickel Plating (Electrodeposited)
SAE AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)

(Copies of these documents are available from www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Material.

3.1.1 Class A carbon steel. Carbon steel washers shall be low carbon steel in accordance with SAE AIR4127.

3.1.2 Class B corrosion-resistant steel. Corrosion-resisting steel washers shall be type 301, 302, 304, 305 or 316 in accordance with SAE AIR4127.

3.1.2.1 Magnetic permeability. Magnetic permeability shall be 2.0 maximum (air = 1.0) for a field strength of H = 200 oersteds when using a magnetic indicator per ASTM A342/A342M.

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3.1.3 Class C - aluminum. Aluminum washers shall be in accordance with ASTM B209/B209M, alloy 1100 or 3003, temper H18 or H28.

3.1.4 Class D - aluminum alloy. Aluminum alloy washers shall be in accordance with SAE AMS-QQ-A-250/5, alclad 2024, temper T4.

3.1.5 Class E - brass. Brass washers shall be in accordance with ASTM B36/B36M alloys UNS C26000, C26800, or C28000 temper H02 or ASTM B121/B121M alloys UNS C33500, C34000, C35000, C35300, or C35600 temper H02.

3.1.6 Class F - copper. Copper washers shall be in accordance with ASTM B152/B152M, temper H04.

3.1.7 Class G - nickel-copper alloy. Nickel-Copper alloy washers shall be in accordance with QQ-N-281, Class A, cold rolled, annealed.

3.1.8 Class H - plastic (nylon). Plastic (nylon) washers shall be in accordance with ASTM D4066, Group 01, Class 1, Grades 1 through 6.

3.1.9 Class J - nickel-copper-aluminum alloy. Nickel-Copper-Aluminum alloy washers shall be in accordance with QQ-N-286, solution annealed and age hardened.

3.1.10 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2 Protective finish. Unless otherwise specified (see 6.2), washers shall be furnished plain, with no protective finish. When finishes are specified, they shall be in accordance with the following paragraphs.

3.2.1 Class A - carbon steel washers.

3.2.1.1 Cadmium plating (Obsolete for New Design, see 6.3). Cadmium /1 plating shall be in accordance with SAE AMS-QQ-P-416; Type II, Class 3 or SAE AMS-C-81562, Type II, Class 3.

/1 Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc-Nickel plate per MIL-PRF-32647, Type K16 or SAE AMS2461, Type II, Grade B. Class 3 are suitable replacements with design authority approval.

3.2.1.2 Zinc-Nickel plating. Zinc-Nickel plating shall be in accordance with MIL-PRF-32647, Type K16 or SAE AMS2461, Type II, Grade B, Class 3.

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3.2.1.3 Zinc plating. Zinc /2 plating shall be in accordance with ASTM B633, Type II or SAE AMS-C-81562, Type II. The thickness on washers for Bolt sizes thru 0.250 inch shall be Class 3 and on washers for bolt sizes over 0.250 inch shall be Class 2.

/2 Type II passivation per ASTM B633 and SAE AMS-C-81562 contains hexavalent chromium which is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. Zinc with hexavalent chromium-free passivation per ASTM F1941/F1941M Fe/Zn 8 or 12 or Zinc-Nickel plate per MIL-PRF-32647, Type K20 are suitable replacements with design authority approval.

3.2.1.4 Zinc plating (hex chrome-free). Zinc plating with hex free passivation shall be in accordance with ASTM F1941/F1941M Fe/Zn 8 or 12 or Zinc-Nickel plate per MIL-PRF-32647, Type K20.

3.2.1.5 Galvanized zinc. Hot-dip galvanizing shall be in accordance with ASTM A153/A153M. The thickness shall be 0.00045 inch minimum.

3.2.1.6 Phosphate coating. Phosphate coating shall be in accordance with MIL-DTL-16232, Type Z, Class 2.

3.2.2 Class B - corrosion-resisting steel washer.

3.2.2.1 Passivation. Passivation treatment shall be in accordance with ASTM A967/A967M, Nitric treatment 2-5, or Citric treatment 1-5, no post cleaning containing hexavalent chromium permitted, or with SAE AMS2700, Method 1, Types 6-8, or Method 2, no post treatment.

3.2.2.2 Black oxide. Black oxide coating shall be in accordance with MIL-DTL-13924, Class 4.

3.2.3 Classes C and D - aluminum and aluminum alloy washers.

3.2.3.1 Anodizing. Anodizing shall be in accordance with MIL-PRF-8625, Type IC, II, or IIB, Class 1. Sealers (clear or any color) containing hexavalent chromium shall be prohibited.

3.2.4 Classes E and F - Brass and copper washers.

3.2.4.1 Tin plating. Tin plating shall be in accordance with ASTM B545 or ASTM B339.

3.2.4.2 Nickel plating. Nickel plating shall be in accordance with SAE AMS-QQ-N-290, Class 1, Grade F.

3.2.4.3 Black chemical. Black chemical finish shall be in accordance with MIL-DTL-495, except hexavalent chromium shall be prohibited.

3.3 Dimensions.

3.3.1 Type A, round washers.

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3.3.1.1 Grade I. Grade I, general assembly purpose washers shall be in accordance with the dimensions in the applicable military specification and ASME B18.21.1.

3.3.1.2 Grade II. Grade II, aeronautical assembly purpose washers shall be in accordance with the dimensions in AIA/NAS NAS1149.

3.3.1.3 Washer faces. Washer faces shall be parallel within 0.002 inch. Washer faces shall be flat within:

0.007 inch for bolt sizes thru 0.500 inch.
0.010 inch for bolt sizes 0.5625 thru 1.250 inches.
0.015 inch for bolts sizes over 1.250 inches.

3.3.1.4 Concentricity. The diameter and the outside diameter shall be concentric within the tolerance of the inside diameter.

3.3.2 Type B square washers. Square washers shall be in accordance with the dimensions in Table I.

3.4 Workmanship. Washers shall be flat and smooth and free from loose scale, sharp edges and all other defects which may affect their serviceability.

Table I. Type B, grade I square washers.

Nom. bolt size	Inside diameter		Outside width		Thickness	
	Min.	Max.	Min.	Max.	Min.	Max.
.375	.438	.469	1.50	1.56	.125	1.56
.4375	.500	.531	1.75	1.81	.125	1.56
.500	.562	.594	2.00	2.06	.188	.219
.625	.719	.750	2.25	2.31	.250	.281
.750	.844	.875	2.50	2.56	.250	.281
.875	.969	1.000	3.00	3.06	.250	.281
1.000	1.094	1.125	3.50	3.56	.375	.406
1.125	1.250	1.312	4.00	4.12	.375	.406
1.250	1.375	1.438	4.50	4.62	.375	.406
1.375	1.500	1.562	5.00	5.12	.375	.406
1.500	1.625	1.688	6.00	6.12	.375	.406
1.750	1.875	1.938	6.50	6.62	.375	.406
2.000	2.125	2.188	7.00	7.12	.375	.406

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4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in this contract or order, the supplier may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this and cited specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all washers of the same type, grade, class, size and protective finish, produced under the same conditions and offered for acceptance at one time.

4.3 Sampling.

4.3.1 Sampling for examination. A random sample of washers shall be selected from each lot in accordance with ASQ Z1.4, inspection level S-2. The Acceptable Quality Level (AQL) defective shall be 4.0 percent defective.

4.3.2 Sampling for tests. The sampling method shall be in accordance with MIL-STD-1916 or ASQ H1607, table 1 or a comparable zero based sampling plan unless otherwise specified by the contract. If the applicable drawing, specification, standard, or quality assurance provision (QAP) specifies Critical, Major and/or Minor attributes, they shall be assigned Verification Levels of VII, IV and II or AQLs of 0.1, 1.0 and 4.0 respectively. Unspecified attributes shall be considered as Major unless sampling plans are specified in applicable documents. For MIL-STD-1916 the manufacturer may use the attribute or variable inspection method at their option. ASQ Z1.4 may be used to set sample lot size but acceptance is at zero non-conformances in the sample lot unless otherwise specified in the contract.

4.4 Examination. Samples selected in accordance with 4.3.1 shall visually examined for material (3.1), protective finish (3.2), dimensions (3.3), and workmanship (3.4). Corrosion resistant steel washers shall be checked for magnetic permeability (3.1.2.1). Any washer containing one or more defects shall be rejected, and if the number of defective washers exceed the acceptance number, the lot represented by the sample shall be rejected.

4.5 Tests.

4.5.1 Protective finish. Tests for protective finishes shall be in accordance with the applicable specification (see 3.2).

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5 PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Flat washers are intended for automotive, aircraft and general bolt, nut and screw applications. Their purpose is to provide an increased bearing surface and to prevent galling of the base material.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Type, grade and class (see 1.2).
c. Protective finish, if required (see 3.2).
d. Dimensions (see 3.3)
e. Selection of applicable levels of packaging and packing (see 5.1).
f. Applicable MS or AN part number.

6.3 Military Procurement. Zinc-Nickel plated washers should be used for replacement of cadmium parts. Items procured under this specification are for military use. It is recommended the design as specified herein be evaluated for use. Cadmium or Hexavalent-Chromium should only be used in aerospace applications, as specified.

6.4 Subject term (key word) listing.

Protective finish
Cadmium
Hexavalent-Chromium
Concentricity
Zinc-Nickel

6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - OS
Air Force - 71
DLA - GS

Review Activities:
Army - AV, CE
Navy - AS, SH, YD

Preparing Activity:
Army - AT

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